

3D SURROUND-VIEW LIDAR · VF SERIES

48 LINES OLEI VF48-50 NO BLIND SPOT

A $360^\circ \times 50^\circ$ surround-view scanner in a $58 \times 55 \times 65$ mm body. Forty-eight lines covering 7 500 square metres to a 50-metre radius, at ± 2 cm and under eight watts.



■ WHY THIS SENSOR

- **$360^\circ \times 50^\circ$ field of view** — surround view and blind-spot coverage from a single sensor position
- **48 lines at $0.36^\circ \times 1.04^\circ$** — dense enough to classify what the platform is approaching, not just detect it
- **50 m radius, 7 500 m² of coverage** — and 30 m at 10% reflectivity for dark targets
- **± 2 cm accuracy at one sigma** — 10 Hz frame rate, 905 nm, Class 1 eye safe
- **$58 \times 55 \times 65$ mm** — the smallest multi-line sensor in the OLEI line — fits where a 97 mm can will not
- **< 8 W across 9–27 V DC** — a wide input range that tolerates unregulated platform power
- **IP67 sealed** — the highest ingress rating in the OLEI scanning line
- **UDP point-cloud output** — 3D point clouds with angle and depth for direct ingestion

50 m

MAX DETECTION RANGE

48

LINES

$360^\circ \times 50^\circ$

FIELD OF VIEW

IP67

SEALED

■ APPLICATION SCENARIOS



AMR & AGV

Full-circle obstacle avoidance from one chassis mount



UNMANNED FORKLIFTS

Pallet and obstacle perception at low speed



UAV & AERIAL

Obstacle avoidance and terrain capture from the air



INTRALOGISTICS

Forklifts, tuggers and conveyor zones in active aisles

PARAMETER	TECHNICAL DATA
Maximum detection range	50 m
Number of lines	48
Field of view	360° × 50°
Angular resolution (H × V)	0.36° × 1.04°
Frame rate	10 Hz
Range at 10% reflectivity	30 m
Accuracy (1 sigma)	±2 cm
Coverage area	Up to 7 500 m ²

PARAMETER	TECHNICAL DATA
Wavelength	905 nm
Eye safety	Class 1
Operating temperature	-20 to 60 °C
Voltage	9–27 V
Power consumption	< 8 W
Protection rating	IP67
Mechanical dimensions	58 × 55 × 65 mm
Data output format	UDP

■ SIZING THE SENSING ENVELOPE

HORIZONTAL SPACING 0.36° is about 31 cm between points at 50 m, 6 cm at 10 m

DARK TARGETS Plan detection from the 30 m @ 10% figure, not the 50 m headline

VERTICAL APERTURE 50° total; a recessed mount truncates this before it truncates the 360°

MOUNTING HEIGHT Set from where the 50° aperture must intersect the floor plane ahead of the platform

VERTICAL SPACING 1.04° is about 91 cm between lines at 50 m, 18 cm at 10 m

ACCURACY ±2 cm at one sigma — allow a multiple of that for clearances

FRAME RATE Fixed 10 Hz — 10 cm of travel per frame at 1 m/s

POWER 9–27 V at under 8 W; no minimum-load or inrush figure is published

Certification. Class 1 eye safety per the manufacturer's specification. Confirm the current laser-safety standard reference, EMC and regional approval status with MorpheusTEK before design-in — the manufacturer's datasheet for this product does not cite the IEC 60825 edition or list EMC, vibration or shock standards.

Functional safety. The manufacturer's specification for this product states no IEC 61496 Type rating and no ISO 13849 performance level. Treat its output as perception and measurement data, not as a rated safety function, and do not use it as the sole protective device. Where a rated scanner is required, ask MorpheusTEK about the OLEI GS1-5.

Notes. The manufacturer's table labels the 30 m figure "Range Accuracy (@10% reflectivity)"; it is a range, not an accuracy — accuracy is the separate ±2 cm (1 sigma) line. Size dark-target performance from 30 m, not 50 m. The 7 500 m² coverage figure follows from the 50 m radius and the 50° vertical aperture; it is not an independently specified parameter. This datasheet has no manufacturer document number.

Installation. Mount the sensor so the full 360° × 50° volume is clear of the platform's own structure; a recessed mount truncates the vertical aperture first and the surround-view claim with it. The 9–27 V input tolerates unregulated platform power, but keep the supply clear of motor transients. Keep the optical window clean.

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